Compression Creep of 63Sn–37Pb Solder at 120 °C: Creep Curves, Minimum Creep Rate and the Stress Exponent

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Course: UMT312L — Mechanical Testing Laboratory

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Date of experiment: 06/11/2026 · Date of submission: 20/11/2026

1 Objective

To study the creep behaviour of a 63Sn–37Pb solder alloy in uniaxial compression at 120 °C under two sustained stresses, to construct the engineering and true creep curves, to extract the minimum creep rate from each, and to determine the stress exponent N of the alloy at that temperature.

2 Theory

2.1 Why solder creeps at 120 °C

Creep is time-dependent plastic deformation under a stress held constant, and it becomes significant above about $0.4\,T_m$ on the absolute scale. For the 63Sn–37Pb alloy used here the melting temperature recorded in the laboratory is 212 °C = 485 K, so a test at 120 °C = 393 K sits at

$$\frac{T}{T_m} = \frac{393}{485} = 0.81, \tag{1}$$

twice the threshold. Solder creeps readily at room temperature for the same reason, which is why creep is the dominant failure mode of solder joints in service.

2.2 The three stages

On loading, a specimen first deforms instantaneously and elastically. Thereafter the strain accumulates in three stages. In the *primary* stage the rate falls continuously: the material strain-hardens as dislocation density rises. In the *secondary* or steady-state stage, hardening is balanced by thermally activated recovery, the dislocation density becomes constant, and the rate reaches a minimum which is nearly constant. In the *tertiary* stage the rate accelerates to failure. In tension this acceleration is mostly geometric — necking raises the true stress. In compression it cannot be, because the section grows rather than shrinks and the true stress *falls*; an accelerating rate in compression therefore signals genuine structural softening, such as dynamic recrystallisation or grain growth.

2.3 Stress dependence

The minimum creep rate follows the Mukherjee–Bird–Dorn relation

$$\dot{\varepsilon}_{\min} \;\propto\; d^{-p}\,\exp\!\left(-\frac{Q_c}{RT}\right)\sigma^{N}, \tag{2}$$

so at fixed temperature and grain size $\dot{\varepsilon}_{\min} \propto \sigma^{N}$, and N follows from two tests at different stresses:

$$N = \frac{\ln(\dot{\varepsilon}_1/\dot{\varepsilon}_2)}{\ln(\sigma_1/\sigma_2)}. \tag{3}$$

The value of N identifies the mechanism: $N \approx 1$ for diffusional flow, $N \approx 2$ for grain-boundary sliding, $N \approx 3$ for viscous glide, and $N \approx 4$–7 for dislocation climb.

2.4 True stress and true strain in compression

The specimen shortens, so engineering strain understates the deformation and the cross-section grows. Taking the volume as constant, with h_0 the original height and Δh the measured shortening:

$$\varepsilon_{\mathrm{eng}} = \frac{\Delta h}{h_0}, \qquad \varepsilon_{\mathrm{true}} = -\ln\!\left(1-\frac{\Delta h}{h_0}\right), \tag{4}$$

$$A = \frac{A_0 h_0}{h} = \frac{A_0}{1-\Delta h/h_0}, \qquad \sigma_{\mathrm{true}} = \frac{F}{A} = \sigma_0\left(1-\frac{\Delta h}{h_0}\right). \tag{5}$$

The true stress therefore falls steadily through a compression creep test, and the nominal stress quoted for a test is only its starting value.

3 Apparatus and Specimens

- Dead-weight compression creep frame with a split furnace, PID temperature control, a load cell and an LVDT reading the shortening of the specimen.
- Data acquisition at 1 Hz, logging elapsed time, LVDT displacement and load.
- Specimens: cylinders of 63Sn–37Pb solder, measured with a vernier at three heights.

Table 1: Specimen geometry and loading, as measured on 6 November 2026.

	Specimen 1	**Specimen 2**
Diameters at three heights (mm)	7.94, 7.92, 7.96	7.98, 7.98, 7.98
Initial height h_0 (mm)	11.75	12.44
Aspect ratio h_0/d	1.48	1.56
Cross-sectional area A_0 (mm^2)	49.265	50.000
Nominal stress (MPa)	8	4
Dead load applied (kg)	39.359	19.657
Dead load (N)	386.1	192.8
Actual initial stress σ_0 (MPa)	**7.837**	**3.857**
Test temperature (°C)	120	120
Duration analysed	76 min	15.08 h

Both aspect ratios fall in the 1.5–2 band the procedure specifies, which keeps end friction from barrelling the specimen badly enough to invalidate the uniaxial assumption. The loads were set from the nominal stresses and the measured areas; because the dead weights are discrete, the stresses actually applied are 7.837 and 3.857 MPa, and those are the values used throughout this report rather than the nominal 8 and 4.

4 Procedure

1. Each specimen was measured with a vernier: the diameter at three heights and the overall height. The mean diameter gave A_0 and hence the dead load needed for the target stress.
2. The LVDT was zeroed and the load cell tared. A 1 kg preload was applied to remove any gap between the compression platens.
3. The specimen was centred on the lower platen, the thermocouples set beside it, the furnace closed and packed with glass wool at both ends.
4. The furnace was brought to 120 °C and held until the LVDT reading stopped drifting, so that the displacement recorded afterwards is creep and not thermal expansion.
5. Acquisition was started, the dead load applied, and the test run until the specimen had accumulated sufficient strain.

5 Treatment of the Record

Two features of the raw data govern how it has been reduced, and both are stated here rather than left implicit.

The instantaneous deformation is not in the record. Both files begin at exactly 0.000 mm and rise smoothly; the largest single-second step anywhere in the 8 MPa record is 4 μm, where a load of 386 N arriving on a solder cylinder would move the LVDT by far more in one second. The transducer was therefore zeroed with the load already resting on the specimen. The elastic and immediate plastic deformation on loading is consequently absent, and the record begins at the onset of creep. *Total strain and creep strain are therefore identical throughout this report*, and no instantaneous component has been subtracted; h_0 is taken as the height at the instant the displacement reads zero.

The 8 MPa test was stopped by a travel limit. Its rate accelerates through tertiary to 1.2×10^{-4} s^{-1}, then at $t = 4576$ s and $\Delta h = 2.990$ mm it collapses by a factor of six and the trace flattens at exactly 3.000 mm for a further 1667 s. That is the frame against a stop, not creep. All analysis of that test is therefore restricted to $t \le 4576$ s; the final 27 % of the file is excluded. The 4 MPa test shows no such knee and is used in full.

6 Results

6.1 Creep curves

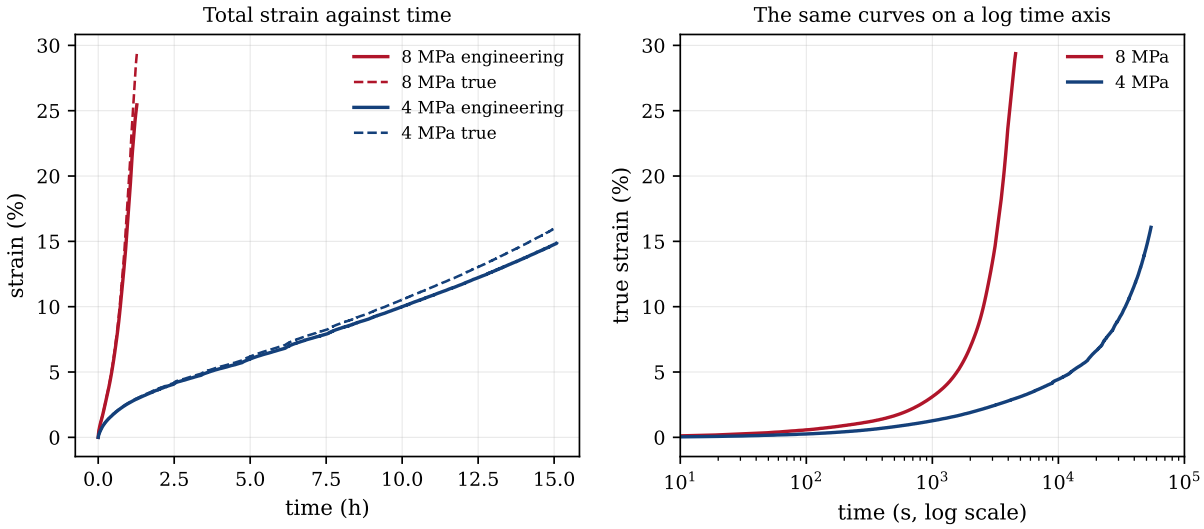


Figure 1: Left: engineering strain (solid) and true strain (dashed) against time for both specimens. Right: the true-strain curves on a logarithmic time axis, which separates the primary stage from the rest.

Table 2: Strain accumulated over the analysed portion of each test.

	7.837 MPa	**3.857 MPa**
Duration	4576 s (76 min)	54 285 s (15.08 h)
Shortening Δh (mm)	2.990	1.847
Engineering strain (%)	25.45	14.85
True strain (%)	29.37	16.07

The two curves have different shapes, and the difference is the point of running two stresses. At 3.857 MPa the specimen passes through a long, clearly defined secondary stage occupying most of fifteen hours. At 7.837 MPa the primary stage is over within about fifteen minutes and the specimen spends most of the test accelerating through tertiary; it reaches in 76 minutes a strain the low-stress test needed fifteen hours to approach. Doubling the stress shortens the life by more than an order of magnitude.

6.2 Creep rate and the minimum

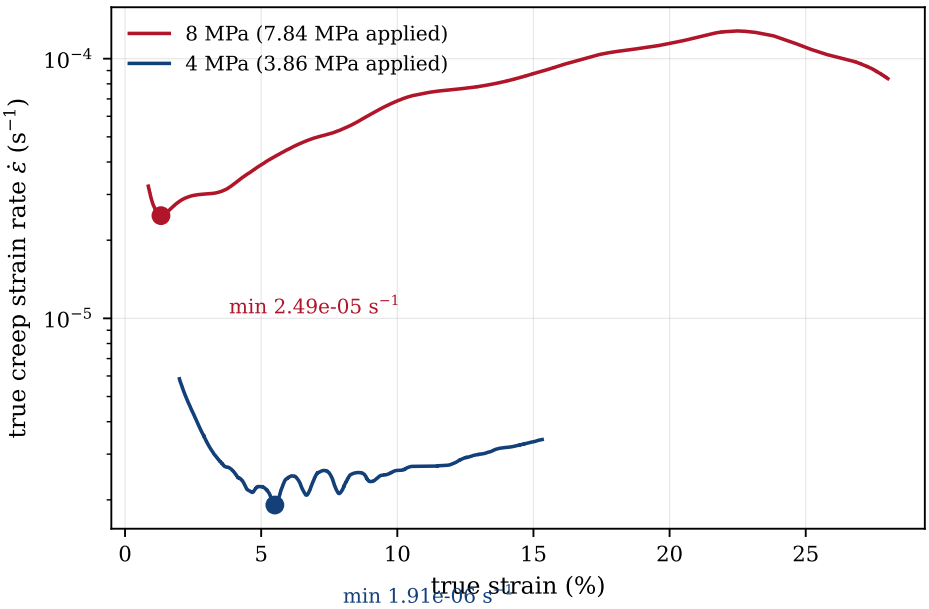


Figure 2: True creep strain rate against true creep strain, obtained by fitting a straight line to the true-strain record over a sliding window and taking its slope. The marked points are the minima.

Both curves fall steeply, pass through a minimum and rise again — the three stages seen as a rate rather than as a strain. The minima are

$$\dot{\varepsilon}_{\min} = 2.49 \times 10^{-5}\ \mathrm{s}^{-1} \qquad \text{at } 1.3\,\% \text{ true strain} \qquad (7.837\ \mathrm{MPa}), \tag{6}$$

$$\dot{\varepsilon}_{\min} = 1.91 \times 10^{-6}\ \mathrm{s}^{-1} \qquad \text{at } 5.5\,\% \text{ true strain} \qquad (3.857\ \mathrm{MPa}). \tag{7}$$

A factor of two in stress produces a factor of thirteen in minimum creep rate.

6.3 Variation of true stress during the test

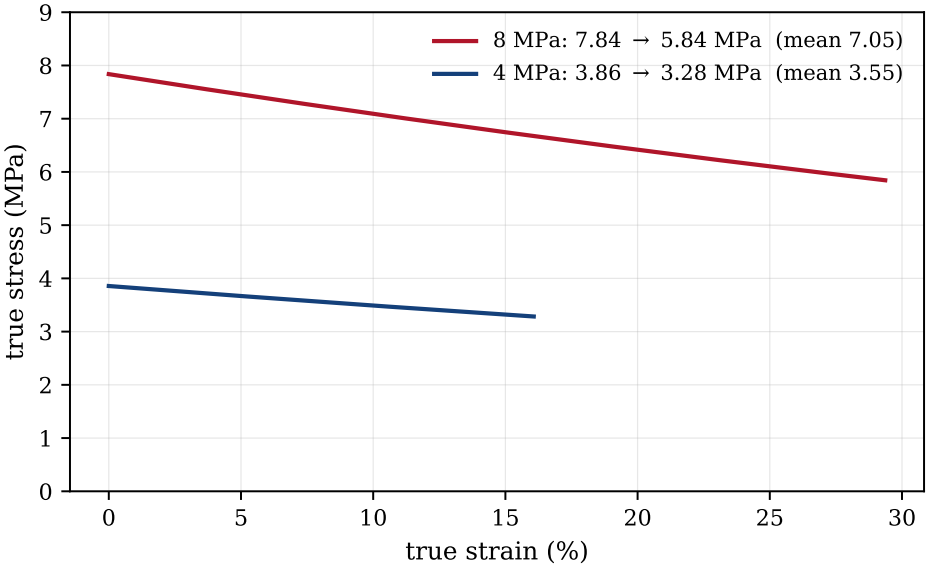


Figure 3: True stress against true strain. Because the specimen shortens and spreads, the true stress falls monotonically from its initial value.

Table 3: Drift of true stress across each test, from Equation 5.

	initial (MPa)	final (MPa)	**mean (MPa)**
Specimen 1	7.837	5.843	**7.049**
Specimen 2	3.857	3.284	**3.545**

The true stress falls by 25 % in the high-stress test and by 15 % in the low-stress one. This matters twice over. First, the average true stress is 10–12 % below the nominal figure, so quoting a creep rate against "8 MPa" overstates the stress that produced it. Second, it disposes of the simplest explanation for tertiary creep: in compression the true stress is *falling* while the rate accelerates, so the acceleration cannot be a stress effect and must come from the structure of the material itself.

6.4 Stress exponent

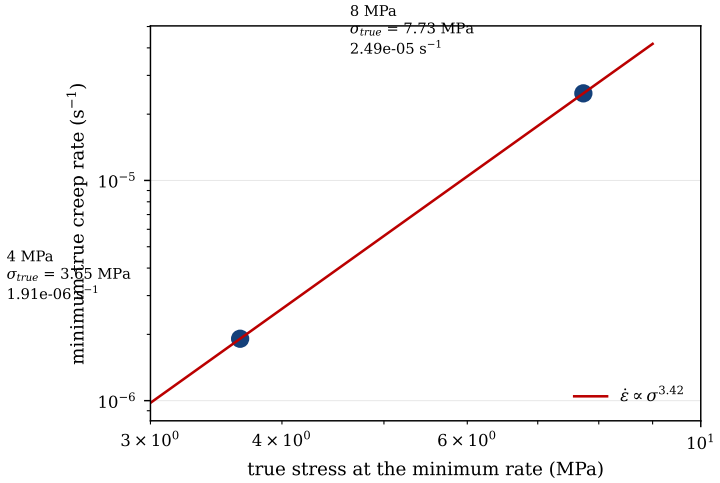


Figure 4: Minimum true creep rate against the true stress acting at that moment, on logarithmic axes. The slope is the stress exponent.

Evaluating Equation 3 with the true stress acting at each minimum — 7.735 and 3.650 MPa — rather than the nominal values:

$$N = \frac{\ln\!\left(2.49 \times 10^{-5} / 1.91 \times 10^{-6}\right)}{\ln(7.735/3.650)} = \frac{2.566}{0.751} = \mathbf{3.42}. \tag{8}$$

A value near 3.4 places this alloy between viscous glide ($N \approx 3$) and dislocation climb ($N \approx 4$–5), which is where eutectic Sn–Pb at $0.8\,T_m$ and a few MPa is normally found. It is not in the diffusional-flow regime, which would require $N \approx 1$ and stresses an order of magnitude lower.

The figure is based on two points, so it fixes the slope without testing whether the relation is actually a power law. Confirming that needs the other stresses tested across the class.

7 Discussion

The experiment separates three ideas that the single phrase "creep rate" runs together.

First, the *rate depends on stress very strongly*. Doubling the stress multiplied the minimum creep rate by thirteen. That is the practical content of $N = 3.42$: in a solder joint, a modest increase in load shortens life out of all proportion.

Second, *the stress itself is not constant.* The test is run at constant load, not constant stress, and in compression the growing section makes the true stress fall by up to a quarter across the test. Any creep rate quoted without saying which stress it belongs to is ambiguous by that much, which is why the stress exponent here is computed from the true stress acting at the instant of the minimum.

Third, *tertiary creep in compression is not geometric.* In a tensile test the accelerating third stage is largely explained by necking. Here the opposite happens — the load is spread over a growing area and the true stress drops — yet the rate still rises by a factor of five in the high-stress test. The acceleration must therefore come from the microstructure. At $0.81\,T_m$ with a fine two-phase eutectic structure, the likely process is continuous recrystallisation and coarsening of the Sn-rich and Pb-rich phases, which softens the alloy as strain accumulates.

A limitation should be recorded. The laboratory note for this set-up states that it resolves displacement up to about 2 mm, intended to cover the primary stage. Both tests exceeded that: the 4 MPa specimen reached 1.85 mm and the 8 MPa specimen 2.99 mm before the travel limit stopped it. The curves obtained are therefore fuller than the set-up is nominally specified for, and the tertiary portion of the 8 MPa test in particular should be read as indicative. The minima, which both occur below 0.7 mm of shortening, are well inside the specified range and are unaffected.

8 Conclusions

1. 63Sn–37Pb solder tested in compression at 120 °C ($T/T_m = 0.81$) showed all three stages of creep at both stresses.
2. At 7.837 MPa the specimen reached 29.4 % true strain in 76 minutes; at 3.857 MPa it reached 16.1 % in 15.1 hours.
3. The minimum true creep rates are 2.49×10^{-5} s^{-1} and 1.91×10^{-6} s^{-1} respectively — a factor of thirteen for a factor of two in stress.
4. The true stress falls through each test, from 7.837 to 5.843 MPa (mean 7.049) and from 3.857 to 3.284 MPa (mean 3.545).
5. The stress exponent, evaluated at the true stress acting at each minimum, is $N = \mathbf{3.42}$, indicating glide- and climb-controlled dislocation creep rather than diffusional flow.
6. Tertiary creep occurred while the true stress was falling, so it reflects softening of the microstructure and not a rising stress.

9 Note on the across-group activity

This report covers the individual group activity in full. The comparison across groups — creep curves at a common temperature for every stress tested, the minimum creep rate against true stress for both temperatures, and the activation energy Q_c from the rate at fixed stress against $1/T$ — will be submitted as a separate report for this experiment once the class data has been collected. Q_c cannot be obtained from the present data at all, since both tests here were run at 120 °C; it requires the groups that tested at 110 °C.